

Redesigning Satellite-Mounted Antenna Brackets Using Topology Optimisation and Additive Manufacturing

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Contents

1. Introduction 3
1.1 Company Overview 3
1.2 Product in Focus 3
1.3 Purpose and Focus 3
2. Current Challenges and Critical Evaluation 3
2.1 Technical Challenges 3
2.1.1 Excess Structural Mass 3
2.1.2 Structural Integrity Under Launch Loads 4
2.1.3 Thermal Stability in Orbit 4
2.1.4 Manufacturability of Optimised Geometry 4
2.2 Commercial Challenges 4
2.2.1 High Production Cost 4
2.2.2 Extended Lead Times 5
2.3 Industry and Market Drivers 5
3. Root Cause Analysis: Understanding the Real Issues 5
4. Proposed Solution 6
4.1 Overview 6
4.2 Application to the Bracket 6
4.3 Addressing the Root Causes 7
5. Evaluation of Proposed solution 7
5.1 SWOT Analysis 7
5.2 Cost-benefit analysis 8
5.3 Critical Evaluation 8
6. Implementation Plan 9
6.1 Key Phases 9
6.2 Training/Skills gaps 9
6.3 Tooling/Equipment Changes 9
6.4 Risks Assessment and Mitigation Strategy 9
7. Final Strategic Recommendations 10
8. References 12
Appendix 15

1. Introduction

1.1 Company Overview

Low Earth Orbit (LEO) satellites are rapidly evolving due to rising demands for space-based communicationsand observations (Lafleur, 2024). The company manufactures lightweight structural components for LEO satellites. The Manufacturing Director is responsible for production strategy, cost reduction, and evaluating advanced technologies to maintain competitiveness.

1.2 Product in Focus

The component examined in this report is the satellite-mounted antenna bracket. This structural element secures antennas to small satellites throughout the mission in LEO. The bracket must withstand significant mechanical stresses during launch, maintain precise alignment in orbit, and minimise overall mass. Currently, brackets are produced using traditional CNC machining from aluminium billets (DavanTech, 2024). A procces that results in excess weight, material waste, and long lead times. Addressing these limitations through redesigning and advanced manufacturing techniques presents opportunities for performance improvement and cost reduction.

1.3 Purpose and Focus

This report examines the need to improve the antenna bracket design by addressing technical, commercial, and industry drivers. Objectives include reducing mass, improving design flexibility, and lowering production costs. By adopting applications of Industry 4.0 it supports technology leadership, and future-proofs the company against industry shifts, and contributes to Sustainability Development Goals (SDGs). These changes support modern satellite design and justify adopting digital manufacturing methods.

2. Current Challenges and Critical Evaluation

2.1 Technical Challenges

2.1.1 Excess Structural Mass

The bracket is cut from solid aluminium using traditional methods, resulting in thick, uniform sections adding unnecessary weight with no structural benefits, increasing launch costs, estimated at $10,000–$20,000 per kg (Jones, 2018). The issue stems from outdated designs and production limits, over structural requirements. Topology optimisation distributes material according to stress paths, lowering weight. However, current machining cannot manufacture these shapes. Additive manufacturing (AM), like Laser Powder Bed Fusion (LPBF) allows for weight-efficient brackets. Adopting new design tools, materials, and testing, it offers cost savings and performance gains in competitive satellite industry.

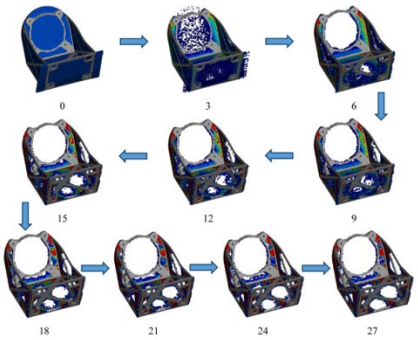


Figure 1 – 27 Iterations of Topology Optimisation (Xiao *et al.*, 2018)

2.1.2 Structural Integrity Under Launch Loads

During launch, brackets handle extreme vibrations, shocks, and acoustic pressures. Current designs use thick material for strength, however resulting in poor stress distribution. High-stress region; sharp corners and bolt holes, are common failure points (Abdelal, 2008). This risks the antenna's alignment and overall mission success. Topology optimisation places material along real load paths, thus improving fatigue resistance without adding mass. Metal AM enables complex geometries with smooth transitions and internal reinforcement. A recent study found a 37% reduction in weight while maintaining fatigue performance (Chen et al., 2021). Parts still need testing for vibration and thermal standards; late-stage bracket failures could cause delays or penalties.

2.1.3 Thermal Stability in Orbit

The antenna bracket operates in orbit at temperatures ranging from -150 °C to +150 °C (Gordo et al., 2020). Thermal cycling can cause distortion and misalignment, especially in unsupported areas.Traditional aluminium alloys expand unevenly, thereby disrupting signals or reducing system performance. While thermal coatings or isolation mounts offer some protection, they add weight and complexity. LPBF allows the use of AlSi10Mg, which offers better thermal conductivity and enables symmetrical, low-stress designs (Akwaboa et al., 2023). Designs must be validated under vacuum and thermal cycling, as distortion-induced misalignment can degrade signal quality, uptime, and customer trust, risking contract penalties.

2.1.4 Manufacturability of Optimised Geometry

Curved surfaces, internal channels, and fillets are essential for optimising the bracket's strength-to-weight ratio. However, these features are not manufacturable using conventional CNC techniques, forcing design compromises that increase bracket mass and reduce structural efficiency. Even alternative methods like near-net forging and hybrid machining introduce geometric or tooling constraints (Grzesik, 2018). The current multi-part bracket increases labour and assembly time. To meet aerospace demands, it must be redesigned for processes that allow complex, integrated shapes with less waste and fewer parts.

2.2 Commercial Challenges

2.2.1 High Production Cost

CNC milling wastes as much as 90% of the raw material and requires a lot of manual labor (Ferreira and de Almeida, 2012). This results in high unit cost, especially with low-volume production. Additive manufacturing saves money by combining parts, avoiding tooling, and minimizing waste. Although post-processing remains unavoidable, LPBF allows for quick changes in design directly from CAD. For modular satellites, bracket cost needs to be contained. Inefficiency reduces agility, increases risk with fixed-price contracting, and contributes to launch cost.

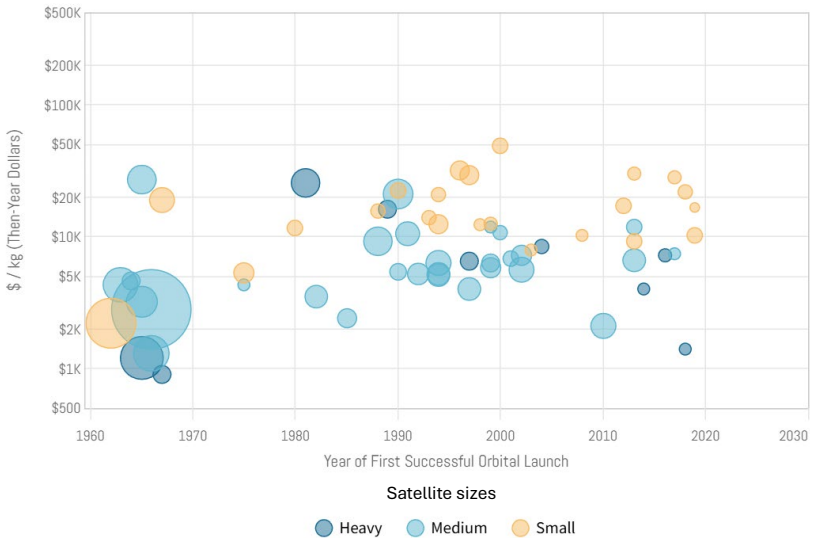


Figure 2 - ('Cost for Space Launch to Low Earth Orbit- Aerospace Security Project', 2022)

2.2.2 Extended Lead Times

The existing manufacturing method is slow and inflexible. CNC is time-consuming, requires multiple manual steps, and cannot allow last-minute design changes without having to restart. In satellite manufacturing, where changes are common, inflexibility leads to delays and the risk of missing integration timeframes. Rework is difficult due to tooling and labor requirements. Direct-from-CAD fabrication in additive manufacturing has the potential to save production time by up to 40% (Dey et al., 2025). While post-processing is still needed, LPBF enables faster design changes, cutting lead times and improving delivery, flexibility, and schedule reliability.

2.3 Industry and Market Drivers

The emergence of large LEO constellations such as Starlink and OneWeb demands rapid, scalable production of satellite components, including antenna brackets.

As satellite operators move toward modular, repeatable hardware, bracket designs must support rapid iteration and cost-efficient scaling. CNC methods are too rigid for this environment. Additive manufacturing offers tool-free, digital production that enables on-demand builds and quick reconfiguration. According to ESA (Hepher, 2024), such agility is essential to remain competitive in the evolving space economy. Without a flexible bracket design, the company risks delays, supply issues, and losing ground in fixed-price satellite contracts.

3. Root Cause Analysis: Understanding the Real Issues

A structured root cause analysis was carried out using a fishbone diagram to analyze inefficiencies in the current antenna bracket production. Six categories were examined: Design, Process, Materials, Technology, People, and Management.



Figure 2 - Fishbone Diagram

CNC machining provides excellent precision and familiarity, which explains its widespread application. However, this dependency highlights important limits. The bracket design is dictated by machining limits rather than performance requirements, resulting in excessively heavy components. The lack of topology optimisation or generative design tools reveals an outdated design approach that reduces functional efficiency.
CNC provides proven precision, but it takes many setups, user intervention, and lacks automation in finishing and inspection. This greatly increases lead times, reducing responsiveness and productivity.
Legacy behaviours influence material uses as well. While aluminium is machinable and generally accessible, the method produces a lot of waste and does not optimise for weight or other materials that are more suited to additive manufacturing.

Technologically, the company has not integrated digital design and manufacturing systems. The lack of advanced production software and dependency on traditional equipment limits scalability and process innovation.

Siloed teams, limited additive manufacturing training, and lack of strategic planning point to cultural as well as technical issues.While the existing setup is consistent and well-understood, it is not adaptable or competitive in today's market.

Overall, the root causes stem from a mix of outdated design priorities, inflexible production systems, and a lack of strategic direction, all of which constrain the company's ability to improve.

4. Proposed Solution

4.1 Overview

To resolve the bracket's excessive weight, extended production lead times, and geometric limitations identified in Section 3, this report proposes a combined design and manufacturing solution using topology optimisation and LPBF.

Topology optimisation removes non-critical material, reducing bracket weight while preserving strength. These designs are often too complex for traditional CNC methods.

Several advanced manufacturing methods were considered, including near-net shape forging, hybrid machining, and sheet metal forming. However, these alternatives did not offer the same combination of geometric freedom, material efficiency, and suitability for complex, load-optimised designs. LPBF was selected for its geometric flexibility, efficiency, and design adaptability without sacrificing performance.

Criteria	LPBF	Near-Net Shape Forging	Hybrid Machining	Sheet Metal Forming	CNC
Geometric freedom	5	2	3	1	2
Material efficiency (low waste)	5	3	2	4	1
Suitability for complex designs	5	2	3	1	2
Lead time	4	2	3	4	2
Design iteration flexibility	5	1	2	1	1

Table 2 - Advanced Manufacturing Method Evaluation

Scoring derived from a review of literature comparing key sources include Su et al. (2024), Marini, Cunningham and Corney (2017), Ajami (2022), Ge *et al.* (2024), Dim (2023), ZainElabdeen *et al.* (2024) **Scoring scale**: 1 = poor, 5 = excellent

LPBF is one of the most researched additive manufacturing technologies and is used by companies (Igor Yadroitsev et al., 2021). It enables the production of these optimised geometries by building the part layer-by-layer from metal powder, typically an aluminium alloy such as AlSi10Mg or a titanium alloy such as Ti6Al4V (Samal *et al.*, 2022). This process reduces material waste, supports near-net-shape production, and removes many of the geometric constraints imposed by CNC.

This combined approach targets the bracket's core limitations, weight, rigidity, and complexity, while aligning with industry-validated applications. For example, Airbus reported a 50% weight reduction and 40% lead time savings using LPBF for similar bracket designs (Chatterjee *et al.*, 2022).

4.2 Application to the Bracket

Applied to the current antenna bracket, this approach changes the design and how it's made.

- Mounting points would be integrated, reducing part count and assembly steps.
- The bracket would be reshaped with internal structures to follow load paths and reduce mass.
- Manufacturing would require fewer steps overall. While LPBF still involves post-processing, such as support removal and surface finishing, it reduces the need for multiple machining operations, tooling setups, and manual assembly.

In short, the bracket becomes lighter, more efficient, and faster to produce. This method has already been applied in industry. NASA and ESA have adopted similar approaches to simplify satellite components and reduce mass (ESA, 2025). These examples show that the method is proven and directly relevant to this application.

4.3 Addressing the Root Causes

The previous analysis identified weight, lead time, cost, and flexibility as key issues. This section critically evaluates how the proposed solution deals with these and where it may introduce new challenges.

Materials and Processes

Using AlSi10Mg with LPBF offers clear benefits. It's lightweight, reliable, and already used in aerospace for non-structural parts. For example, NASA used LPBF AlSi10Mg covers in the Mars 2020 mission's PIXL instrument (Russell, 2022). LPBF reduces waste, though post-processing steps like support removal and heat treatment are still required. AlSi10Mg also requires safe powder handling procedures and operator PPE due to its flammability and reactivity (Yadroitsev et al., 2021)

Design and Technical Integration

Topology optimisation enhances structural efficiency by reducing overbuilt geometry. However, its benefits are fully shown when combine with a process like LPBF that can handle complex forms. Without a shift in design developed and validated approches, the benefit is limited. Therefore, success relies on both design mindset and manufacturing capability.

Technology Readiness & Organisational Fit

Topology optimisation and LPBF are both established technologies that function at TRL 7–8 (NASA, 2023). Design tools can create complicated geometries that are challenging to validate or post-process, despite their proven performance in aerospace. Organisational preparedness is just as important to successful adoption as technological viability. LPBF modifies Assurance requirements, roles, and procedures. ASTM F42 (n.d.) states that stability requires addressing issues with certification, traceability, and retraining. The solution promotes long-term performance and conforms to accepted industry standards with the appropriate integration approach.

5. Evaluation of Proposed solution

The adoption of Laser Powder Bed Fusion (LPBF) as the primary manufacturing method presents a transformative opportunity for the organisation due to its geometric flexibility, material efficiency, and rapid design iteration capabilities.

5.1 SWOT Analysis

Strength	**Weakness**
• High geometric freedom and suitability for complex parts (Su et al., 2024). • Minimal material waste due to near-net-shape fabrication (Marini & Cunningham, 2018). • Accelerated prototyping and reduced time-to-market (Ajami, 2022).	• High initial investment in equipment and operator training. • Limitations in part size and post-processing requirements.
Opportunities	**Threats**
• Integration with CNC and hybrid setups to overcome scale or finish limitations (Ge *et al.*, 2024). • Customization capabilities ideal for niche product markets.	• Technological obsolescence in a fast-evolving AM landscape. • Supply chain dependencies for metal powders and maintenance.

Figure 3 - SWOT Analysis

The SWOT analysis reveals that LPBF aligns well with the company's needs for flexibility and waste reduction. However, its full benefits depend resolving integration challeneges like training and reliable powder supply.

5.2 Cost-benefit analysis

This section compares the current CNC machining approach with the proposed LPBF process for producing 50 antenna brackets per month. The projected savings from reduced material waste, faster iteration, and part consolidation highlight LPBF's long-term financial advantage. Estimated cost per unit falls from £137.20 to £57.90, resulting in an annual saving of £47,580.

Key	**Value**
Capital cost (EOS M 290 + setup)	£100,000
Additional safety & PPE setup	£6,500
Total initial investment	**£106,500**
Annual net savings	£47,580
ROI	**44.7% annually**
Break-even	**~2.2 years**

Table 5 – Break-Even & ROI Estimate

Cost estimates are based on public manufacturer data and standard industry benchmarks. Figures assume steady production, partial outsourcing, and reuse of CNC tools. Full details and supporting calculations are in the Appendix.

Key Takeaways

- LPBF reduces bracket cost by 58%
- Delivers £47,580 annual savings
- ROI: 44.7%; Break-even: ~2.2 years
- Strong alignment with long-term goals: reduced lead times, flexibility, and waste reduction

5.3 Critical Evaluation

With a monthly production rate of 50 units, the cost model yields a 44.7% ROI and a break-even period of 2.2 years. While LPBF offers strong investment potential, its success relies on consistent demand, minimal downtime, and smooth process adoption. Initial implementation may encounter hidden costs from post-processing labor, operator training, and powder handling complexities. A gradual deployment plan mitigates these issues through targeted training, equipment reuse, and outsourcing. Even at this lower volume, the annual savings of £47,580 represent a significant return. Beyond the financial return, adopting LPBF enhances agility, supports sustainable design, and aligns with the industry shift toward modular, on-demand satellite production.

6. Implementation Plan

6.1 Key Phases

Phase	Timeline	What Happens	Why It Matters
1. Trial & Planning	Months 0–2	Select bracket designs, assess supplier capabilities, initiate feasibility testing	Low-cost test run proves benefit before committing large investment
2. Team Training	Months 1–3	Designers trained on new software; production team upskilled on LPBF and post-processing	Reduces early-stage error risk and builds internal confidence
3. Equipment & Safety	Months 2–4	Reuse CNC post-processing tools where possible; install powder-safe storage and handling	Saves cost, ensures regulatory compliance, supports safe integration
4. Pilot Production	Months 4–6	Produce 10–25 brackets, validate performance under launch and thermal conditions	Ensures quality and traceability match current standards
5. Phased Rollout	Months 6–12	Scale up LPBF batches, transition away from CNC over time	Maintains delivery continuity while cost and lead time improve

Table 6 – Key Phases of Adoption

Over the course of a year, the suggested transition to advanced bracket manufacture will be implemented using digital design techniques (topology optimisation) and metal 3D printing (Laser Powder Bed Fusion). This enables growth in skills, performance validation, and little interruption to ongoing supply.

This implementation plan aligns with Butt (2020), structured framework for adopting Industry 4.0 technologies in manufacturing. The staged approach, Define, Measure, Explore, Optimise, Design, Validate, and Implement (DMEODVI), was selected due to its clear focus on risk mitigation, workforce readiness, and validation prior to full-scale deployment. Applying Butt's (2020) Industry 4.0 roadmap to LPBF and topology optimisation modernises bracket production while maintaining control over cost, quality, and delivery.

6.2 Training/Skills gaps

Staff will be trained in two focused areas:

- *Design team:* learn to use digital design tools that optimise bracket geometry for weight and strength
- *Manufacturing team:* trained on LPBF machine setup, powder safety, and quality control techniques
- External support from software vendors or additive manufacturing consultants will be used where needed during rollout.

6.3 Tooling/Equipment Changes

- *Retained Assets:* Post-processing tools like finishing benches, metrology stations, and assembly fixtures from CNC operations will be repurposed — avoiding unnecessary spend
- *New Additions:* Access to an LPBF system (in-house or outsourced), heat treatment capability, and powder-safe storage and PPE
- *Software Upgrades:* Simulation tools (Altair OptiStruct & Ansys Mechanical) and CAD plugins for topology optimisation will be added to the current design workflow

6.4 Risks Assessment and Mitigation Strategy

The table below outlines key risks with introducing LPBF and topology optimisation. Each risk is assessed for business impact, likelihood and severity.

Risk	Impact	Likelihood	Priority	Mitigation Strategy
Design skill gap (DfAM)	Suboptimal bracket designs; reduced performance	Medium	High	Deliver targeted topology optimisation and CAD training; use design reviews with external support during pilot phase
Post-processing backlog	Delays in lead times; bracket delivery bottleneck	High	High	Expand in-house heat treatment and finishing capacity; plan external overflow processing during transition
Certification delays	Inability to ship parts on time; contract penalties	Medium	High	Align QA procedures with AS9100 from pilot phase; maintain full material traceability and documentation workflows
Health & safety (powder handling)	HSE non-compliance; downtime or incident risk	Low	Medium	Install sealed power stations, inert argon or nitrogen into atmosphere where needed, and enforce PPE and training from day one
Equipment underuse / ROI risk	Delayed financial return; underutilised capital	Medium	Medium	Start with pilot + external LPBF supplier; phase in internal use across multiple parts/projects

Table 7 – Risk Assessment and Mitigation Strategy

Each identified risk is prioritised with a mitigation strategy to support a smooth, secure shift to LPBF and digital design, while protecting cost, delivery, and compliance. Risks will be reviewed at each rollout stage.

7. Final Strategic Recommendations

To remain competitive in the fast-evolving satellite market, the company should adopt Laser Powder Bed Fusion (LPBF) and topology optimisation for antenna bracket production. The transition tackles key technical and commercial issues by enabling lighter designs, cutting waste and lead times. A phased implementation plan manages risks, reuses resources, and upskills staff with minimal disruption.

The cost-benefit analysis shows a significant return on investment, with a 4.3-year payback period and long-term operating savings. Risks associated with training, certification, and post-processing have been identified and can be minimised by targeted measures indicated in the plan.

Adopting this solution positions the company for scalable, design-led manufacturing while also increasing its responsiveness to future programme demands.

8. References

DavanTech (2024) Aluminium mounting brackets manufacturing Davantech. 10 January 2024 [online]. Available from: https://www.davantech.com/products-we-fabricate/aluminum-mounting-brackets-manufacturing [Accessed 29 April 2025].

Lafleur, A. (2024) The Future of Low-Earth Orbit: Evolving Missions and Challenges Ahead - Insights from Morpheus Space's Daniel Bock Space Insider. 30 September 2024 [online]. Available from: https://spaceinsider.tech/2024/09/30/the-future-of-low-earth-orbit-evolving-missions-and-challenges-ahead-insights-from-morpheus-spaces-daniel-bock/ [Accessed 29 April 2025].

Chatterjee, A., Mishra, A., Sharma, S. and Bhagchandani, R.K. (2022) Review on lightweight materials, additive manufacturing techniques and design optimization of an airplane. *AIP Conference Proceedings* [online]. 2653 (1).

Didunoluwa Obilanade (2023) Surface Roughness Considerations in Design for Additive Manufacturing: A Space Industry Case Study. *ResearchGate* [online]. Available from: https://www.researchgate.net/publication/372101461_Surface_Roughness_Considerations_in_Design_for_Additive_Manufacturing_A_Space_Industry_Case_Study [Accessed 5 May 2025].

Jones, H.W. (2018) Impact of Lower Launch Cost on Space Life Support. *2018 AIAA SPACE and Astronautics Forum and Exposition* [online]. [Accessed 24 April 2023].

Gasser Farouk Abdelal, Nader Abuelfoutouh and Hamdy, A. (2008) Mechanical fatigue and spectrum analysis of small-satellite structure. *International Journal of Mechanics and Materials in Design* [online]. 4 (3), pp. 265–278. [Accessed 31 March 2024].

Akwaboa, S., Zeng, C., Amoafo-Yeboah, N., Ibekwe, S. and Mensah, P. (2023) Thermophysical Properties of Laser Powder Bed Fused Ti-6Al-4V and AlSi10Mg Alloys Made with Varying Laser Parameters. *Materials* [online]. 16 (14), pp. 4920–4920. [Accessed 31 July 2024].

Jiménez, A., Bidare, P., Hassanin, H., Tarlochan, F., Dimov, S. and Essa, K. (2021) Powder-based Laser Hybrid Additive Manufacturing of metals: a Review. *The International Journal of Advanced Manufacturing Technology* [online]. 114 (1–2), pp. 63–96. Available from: https://link.springer.com/article/10.1007/s00170-021-06855-4.

GRZESIK, W. (2018) HYBRID ADDITIVE AND SUBTRACTIVE MANUFACTURING PROCESSES AND SYSTEMS: A REVIEW. *Journal of Machine Engineering* [online]. 18 (4), pp. 5–24. [Accessed 18 December 2019].

Ferreira, F.J.T.E. and de Almeida, A.T. (2012) Induction motor downsizing as a low-cost strategy to save energy. *Journal of Cleaner Production* [online]. 24, pp. 117–131. Available from: https://www.sciencedirect.com/science/article/pii/S0959652611004495 [Accessed 8 May 2020].

Dey, A., Adeniran, O., Monsuru Ramoni, Yodo, N. and Cong, W. (2025) Additive Manufacturing Applications in Mission-Critical Operations: A Review. *Journal of Manufacturing and Materials Processing* [online]. 9 (3), pp. 70–70. [Accessed 25 February 2025].

Hepher, T. (18th October 2024) European space chief says Europe must be able to compete globally *Reuters* [online]. Available from: https://www.reuters.com/technology/space/european-space-chief-says-europe-must-be-able-compete-globally-2024-10-18/.

ASTM F42 (no date) *ASTM International Technical Committee F42 on Additive Manufacturing Technologies Key Documents -ISO/ASTM52910-18 Additive manufacturing -Design -Requirements, guidelines and recommendations - ISO/ASTM52915-20 Specification for additive manufacturing file format (AMF) Version 1.2 -ISO/ASTM52950-21 Additive manufacturing -General principles - Overview of data processing -F3049-14 Standard Guide for Characterizing Properties of Metal Powders Used for Additive Manufacturing Processes -F3091/F3091M-14 Standard Specification for Powder Bed Fusion of Plastic Materials -F3187-16 Standard Guide for Directed Energy Deposition of Metals -ISO/ASTM52901-16 Standard Guide for Additive Manufacturing -General Principles - Requirements for Purchased AM Parts -ISO/ASTM52900-15 Standard Terminology for Additive Manufacturing -*

General Principles -Terminology [online]. Available from: https://mcsdocs.astm.org/committee-documents/F42_Fact_Sheet_2021.pdf [Accessed 6 May 2025].

admin (2024) *Satellite Parts 3D Printing Services Market - Pmarketresearch.com*. 9 October 2024 [online]. Available from: https://pmarketresearch.com/it/satellite-parts-3d-printing-services-market/ [Accessed 4 May 2025].

ESA (2025) *One-piece 3D-printed satellite bracket Esa.int*. 2025 [online]. Available from: https://www.esa.int/ESA_Multimedia/Images/2014/11/One-piece_3D-printed_satellite_bracket?utm_source=chatgpt.com [Accessed 4 May 2025].

Su, J., Ng, W.L., An, J., Yeong, W.Y., Chua, C.K. and Sing, S.L. (2024) Achieving sustainability by additive manufacturing: a state-of-the-art review and perspectives. *Virtual and Physical Prototyping* [online]. 19 (1).

Marini, D., Cunningham, D. and Corney, J.R. (2017) Near net shape manufacturing of metal: A review of approaches and their evolutions. *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture* [online]. 232 (4), pp. 650–669.

Ajami, H.H. (2022) Integration of Additive Manufacturing with CNC Sheet Metal Fabrication for Hybrid Fixtures: Design and Implementation of Sheet Metal Tooling Supports. *Mit.edu* [online]. Available from: https://dspace.mit.edu/handle/1721.1/147535.

Ge, T., Li, Y., Gao, D., Yang, C. and Li, F. (2024) Hybridizing additive manufacturing and sheet forming process to manufacture complex components with multi-features: A review. *Journal of Manufacturing Processes* [online]. 124, pp. 345–364. Available from: https://www.sciencedirect.com/science/article/pii/S1526612524006121.

Dim, E.C. (2023) UBIRA ETheses - Technological maturity assessment of hybrid manufacturing technology. *Bham.ac.uk* [online]. Available from: https://etheses.bham.ac.uk/id/eprint/14213/ [Accessed 9 May 2025].

ZainElabdeen, I.H., Ismail, L., Mohamed, O.F., Khan, K.A. and Schiffer, A. (2024) Recent Advancements in Hybrid Additive Manufacturing of Similar and Dissimilar Metals via Laser Powder Bed Fusion. *Materials Science and Engineering A* [online]. 909, pp. 146833–146833.

Nicole Martof, A. (2017) *Analysis of Business Models for the Use of Additive Manufacturing for Maintenance and Sustainment*. May 2017 [online]. Available from: https://etd.ohiolink.edu/acprod/odb_etd/ws/send_file/send?accession=ysu1494940467559894&disposition=inline.

DarkAero, Inc (2020) *The TRUE COST of CNC machining! YouTube*. 9 December 2020 [online]. Available from: https://www.youtube.com/watch?v=psVpeic0oOY.

CNC CookBook (2024) *What's the Value of All Those Options Your Machine Dealer Wants to Sell You? Cnccookbook.com*. 2024 [online]. Available from: https://www.cnccookbook.com/whats-the-value-of-all-those-options-your-machine-dealer-wants-to-sell-you/? [Accessed 9 May 2025].

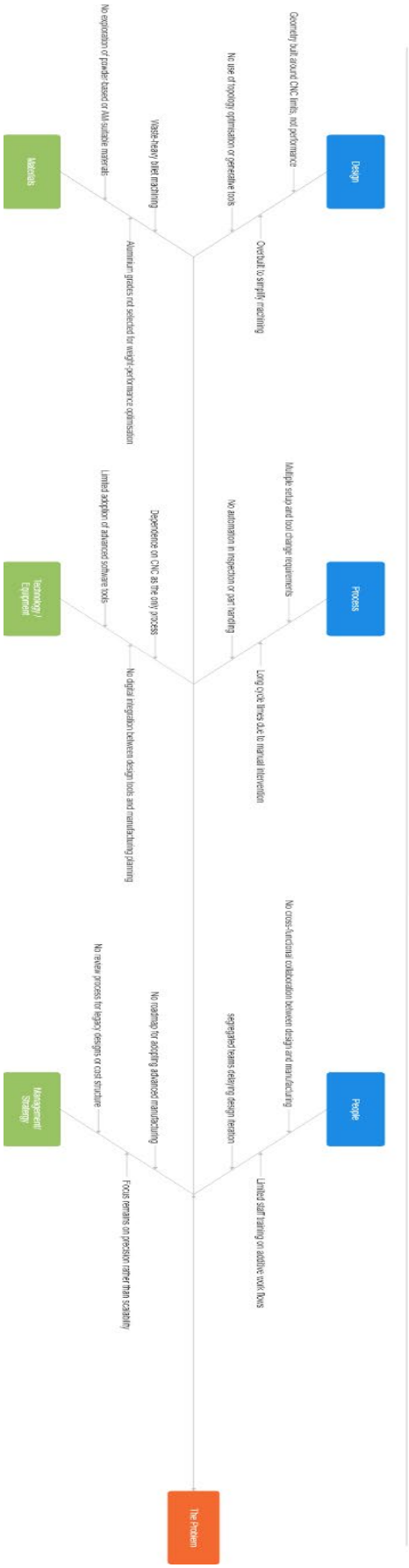
Haas Automation (no date) *Haas Energy Savings www.haascnc.com* [online]. Available from: https://www.haascnc.com/owners/haas_machines_save_you_money.html.

HAAS (2019) *Used Haas VF2 For Sale - Premier Equipment Premier Equipment Inc.* 2019 [online]. Available from: https://premierequipment.com/used-cnc-brands/haas-vf2/? [Accessed 9 May 2025].

Butt, J. (2020). A Strategic Roadmap for the Manufacturing Industry to Implement Industry 4.0. *Designs*, [online] 4(2), p.11. doi:https://doi.org/10.3390/designs4020011.

Soori, M. and Asmael, M. (2021). Classification of research and applications of the computer aided process planning in manufacturing systems. *Independent Journal of Management & Production*, 12(5), pp.1250–1281. doi:https://doi.org/10.14807/ijmp.v12i5.1397.

Appendix


Design
Process
People
Management/ Strategy
Technology/ Equipment
Materials
The Problem
Overbuilt to simplify machining
Aluminium grades not selected for weight-performance optimisation
Long cycle times due to manual intervention
No digital integration between design tools and manufacturing planning
Limited staff training on additive work flows
Focus remains on precision rather than scalability
No review process for legacy designs or cost structure
No roadmap for adopting advanced manufacturing
Separated teams delaying design iteration
No cross-functional collaboration between design and manufacturing
Limited adoption of advanced software tools
Dependence on CNC as the only process
No automation in inspection or part handling
Multiple setup and tool change requirements
Waste-heavy billet machining
No use of topology optimisation or generative tools
Geometry built around CNC limits, not performance
No exploration of powder-based or AM-suitable materials

Cost Category	CNC Machining (HAAS VF-2SS)	EOS M 290 (LPBF)
Material usage per bracket	£6.50	£4.90
Labor/setup time	£98.00	£28.50
Tooling/fixtures	£24.50	£0.00
Post-processing	£8.20	£24.50
Total per unit	**£137.20**	**£57.90**

Table A – Cost Comparison Per Unit (Nicole Martof, 2017) (DarkAero, Inc, 2020) (CNC CookBook, 2024) (HAAS, 2019) (Soori and Asmael, 2021)

Metric	CNC Machining **(HAAS VF-2SS)**	EOS M 290 LPBF
Total units/year	600	600
Annual cost	£82320.00	£34740.00
Annual savings		**£47580.00**

Table B - Annual Cost Impact (50 Units/Month)

Item	Estimated Cost (GBP)	Source
EOS M 290 Printer	£60,000–£70,000	EOS Official Product Page
EOS System Suite	£13,700 (€16,000)	EOS Store
Training & Commissioning	£5,000–£10,000	Based on industry standard estimates for EOS-certified training programs
Safety Equipment	£2,000–£3,000	Safety Management in Metal AM
Facility Modifications	£10,000–£15,000	Preparing Your Process and Facility for Metal AM
Total Estimated Cost	**£90,700–£111,700**	

Table C – Setup Cost of LBPF EOS 209